**Learning Objective 6.1:** I can transmit and receive CW (Morse) using GNU Radio and decode a received CW stream.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **Morse timing.** A dit lasts one unit of $1.2/W$ seconds at $W$ words per minute. A dah lasts 3 units, and the key stays up for 1 unit inside a letter, 3 between letters, and 7 between words. SOS is `... --- ...`.

   (a) Find the length of a dit, a dah, and the gap between words at 15 WPM.

   $$t_{\text{dit}} = \frac{1.2\ \text{s}}{15} = 80\ \text{ms}$$
   $$t_{\text{dah}} = 3 \times 80 = 240\ \text{ms}$$
   $$t_{\text{word}} = 7 \times 80 = 560\ \text{ms}$$

   dit: 80 ms

   dah: 240 ms

   word gap: 560 ms

   (b) Count the units in SOS, including the 7-unit gap after it, and find how long it takes to send at 15 WPM.

   Each S is three dits with two 1-unit gaps, 5 units. O is three dahs with two gaps, 11 units. Two letter gaps add 6 and the word gap 7:

   $$5 + 11 + 5 + 6 + 7 = 34\ \text{units}$$
   $$34 \times 80\ \text{ms} = 2.72\ \text{s}$$

   units: 34

   time: 2.72 s

   (c) A beacon sends its three-letter identifier at 7 WPM. In words, explain why the standard word PARIS lets us state its speed without knowing which letters it sends.

   WPM defines the unit, not the letters: one WPM sends the 50 units of PARIS once a minute, so at 7 WPM every unit lasts $1.2/7 = 171$ ms whatever the letters are. The identifier takes a different number of units than PARIS, so it does not take exactly a seventh of a minute, but its dits, dahs, and gaps all follow from that one unit.

2. **Decoding marks and spaces.** The L12 decoder is set to 15 WPM, so one unit is 80 ms. It calls a mark under 2 units a dit and a longer one a dah, ends a letter when a space reaches 2 units, ends a word when it reaches 5, and ignores any change that lasts less than 0.3 unit. A human operator sends the following, in milliseconds, starting with a mark:

| Mark | 230 | | 255 | | 245 | | 240 | | 75 | | 250 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Space | | 85 | | 70 | | 260 | | 95 | | 80 | |

and then the key stays up for 700 ms.

(a) Find the decoder's boundaries in milliseconds, and decode the text.

A mark or a space of $2 \times 80 = 160$ ms is the dit–dah and in-letter–letter boundary, $5 \times 80 = 400$ ms ends a word, and $0.3 \times 80 = 24$ ms is the noise limit. The marks of 230, 255, and 245 ms are dahs, and the 260 ms space ends the letter: `---` is O. Then 240, 75, and 250 ms read `-.-`, which is K, and the 700 ms space ends the word. The text is OK.

OK

(b) The operator pauses, and the 260 ms space stretches to 420 ms. What does the decoder print now?

420 ms is more than 400 ms, so the decoder ends a word there and prints `O K` with a space between the letters.

O K

(c) During the 700 ms space, a noise burst holds the key down for 15 ms. Explain what the decoder does with it, and what it would do with a 60 ms burst.

15 ms is less than the 24 ms noise limit, so the decoder ignores it and the space continues. A 60 ms burst lasts longer than 24 ms, so the decoder accepts it as a mark under 160 ms, a dit, and if the key then stays up long enough, it prints an extra E.

3. **Key clicks.** Lesson 12 measured the keyer's spectrum at 20 WPM. With raised-cosine edges, the level 1 kHz from the carrier, compared with hard keying, was $-10$ dB for 1 ms edges, $-24$ dB for 2 ms, $-40$ dB for 5 ms, and $-52$ dB for 10 ms.

   (a) A club station must keep its keying at least 30 dB below hard keying, 1 kHz from the carrier. Which of the four edge times meet that, and which is the shortest that does?

   The 5 ms and 10 ms edges meet it, at $-40$ and $-52$ dB. The 2 ms edge reaches only $-24$ dB. The shortest that meets the limit is 5 ms.

   | 5 ms |
   |---|

   (b) An operator wants 10 ms edges at 60 WPM. Find the dit length, and how much of each dit stays at full power. Each edge takes the full edge time, half of it before the nominal edge and half after.

   $$t_{\text{dit}} = \frac{1.2\text{ s}}{60} = 20\text{ ms}$$
   $$20 - 10 = 10\text{ ms}$$

   | dit: | 20 ms |
   |---|---|
   | full: | 10 ms |

   Half of each dit goes to the rising and falling edges.

   (c) In words, explain who a hard-keyed transmitter bothers and why, and why the operator in part (b) should choose a shorter edge.

   Hard keying spreads energy kilohertz away from the carrier, so a receiver tuned to another station nearby hears a click at every edge of the hard-keyed signal. The station itself decodes the same either way. At 60 WPM a 10 ms edge leaves only 10 ms of each 20 ms dit at full power, so the dits turn into rounded bumps and a receiver in noise has less of each mark to detect. A 5 ms edge keeps 15 ms of each dit at full power and still keeps the clicks well down.

4. **The product detector and the CW filter.** The L12 receiver moves the tuned frequency to 0 Hz, multiplies by a 700 Hz BFO, and keeps the real part. Its CW filter is a band-pass centered on 700 Hz. The noise has amplitude $\sigma$, spread over 32 kHz, and the keyed carrier has power 1, so the SNR in a filter of width $B$ is $32{,}000/(\sigma^2 B)$.

   (a) A cadet tunes the receiver 120 Hz above the carrier. What pitch does the tone have, and does it pass a 200 Hz CW filter? A 1 kHz filter?

   Tuned 120 Hz high, the carrier lands at $-120$ Hz, so the tone is $700 - 120 = 580$ Hz. The 200 Hz filter passes 600 to 800 Hz, so the tone falls outside it and the decoder loses the signal. The 1 kHz filter passes 200 to 1200 Hz, so the tone gets through.

   tone: 580 Hz

   200 Hz: no

   1 kHz: yes

   (b) At $\sigma = 8$, find the SNR in the 200 Hz filter and in a 50 Hz filter.

$$\text{SNR}_{200} = \frac{32{,}000}{64 \times 200} = 2.5 = 4.0 \text{ dB}$$

$$\text{SNR}_{50} = \frac{32{,}000}{64 \times 50} = 10 = 10.0 \text{ dB}$$

   200 Hz: 4.0 dB

   50 Hz: 10.0 dB

   (c) Lesson 12's 200 Hz filter decoded without errors down to 10 dB. Explain in words why the 50 Hz filter in part (b) is the better choice in this noise, and what the receiver gives up by using it.

   The signal is only about 100 Hz wide, so most of what a wider filter passes is noise. Cutting the width by four cuts the noise by four, 6 dB, which brings the SNR to the 10 dB where the 200 Hz filter decoded cleanly. What the receiver gives up is tuning range: the tone must stay inside the narrow filter, so the receiver must tune within 25 Hz of the carrier, a quarter of the 200 Hz filter's range, and transmitter drift matters four times as much.

5. **Setting the threshold.** In the L12 receiver the Threshold block switches the key on when the envelope rises above $1.2\times$ `thr` and off when it falls below $0.8\times$ `thr`. On a noisy link, the envelope plot shows a space level of about 0.30 and a mark level of about 0.70.

   (a) Choose `thr`, and find the two levels at which the key switches.

   Halfway between the levels:

   $$\texttt{thr} = \frac{0.30 + 0.70}{2} = 0.50$$

   $$\text{on} = 1.2 \times 0.50 = 0.60, \quad \text{off} = 0.8 \times 0.50 = 0.40$$

   thr: 0.50

   on / off: 0.60 / 0.40

   (b) A teammate leaves `thr` at 0.3, the value that worked with little noise. Explain what the key and the decoded text do.

   The key switches on at 0.36, barely above the 0.30 space level, so noise peaks during the spaces turn it on. The decoder sees short extra marks, prints extra Es and Ts, and splits or corrupts letters, so most of the text comes out wrong.

   (c) Another teammate sets `thr` to 0.8. Explain what happens, and why the hysteresis does not help.

   The key now switches on only above 0.96, well above the 0.70 mark level, so only the strongest noise peaks during marks turn it on. The key misses most marks, and the decoder prints few letters, mostly short ones. Hysteresis only stops chatter near the threshold; it cannot fix a threshold that sits outside the range between the two levels.

**Documentation:** _____